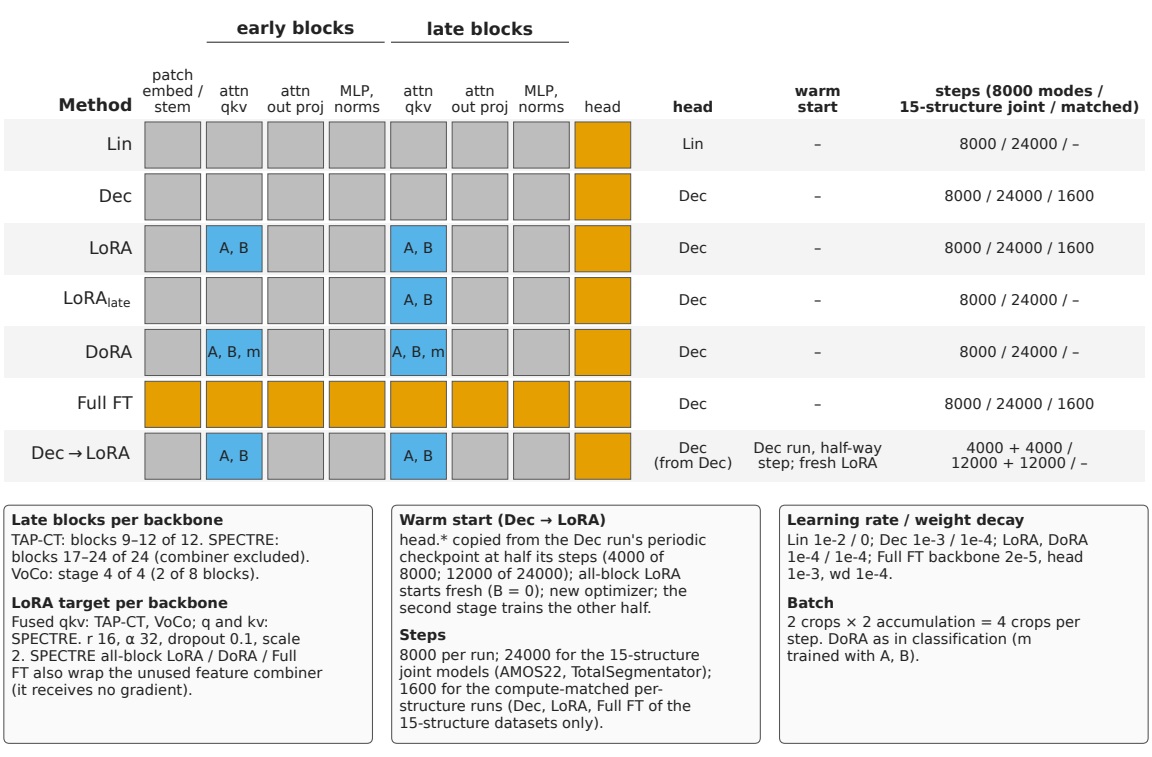

Late blocks per backbone
TAP-CT: blocks 9–12 of 12. SPECTRE: blocks 17–24 of 24 (combiner excluded). VoCo: stage 4 of 4 (2 of 8 blocks).

LoRA target per backbone
Fused qkv: TAP-CT, VoCo; q and kv: SPECTRE. r 16, α 32, dropout 0.1, scale 2. SPECTRE all-block LoRA / DoRA / Full FT also wrap the unused feature combiner (it receives no gradient).

Warm start (Dec → LoRA)
head.* copied from the Dec run's periodic checkpoint at half its steps (4000 of 8000; 12000 of 24000); all-block LoRA starts fresh (B = 0); new optimizer; the second stage trains the other half.

Steps
8000 per run; 24000 for the 15-structure joint models (AMOS22, TotalSegmentator); 1600 for the compute-matched per-structure runs (Dec, LoRA, Full FT of the 15-structure datasets only).

Learning rate / weight decay
Lin 1e-2 / 0; Dec 1e-3 / 1e-4; LoRA, DoRA 1e-4 / 1e-4; Full FT backbone 2e-5, head 1e-3, wd 1e-4.

Batch
2 crops × 2 accumulation = 4 crops per step. DoRA as in classification (m trained with A, B).

frozen (pretrained weights) low-rank adapter, trained trained in full